

FLIGHT TEST CARD

Basic informations

Test Number: FTC-001

Date: 12/13/2024

Aircraft Type/Model: Cessna Citation C560 XLS

Test Crew:

- Pilot: [_____]

- Test Engineer: [Simone Muscolino]

Objectives

1. Evaluate flight performance as a function of:
 - I. Altitude;
 - II. Flight Mach;
 - III. Oswald factor;
 - IV. Bank angle φ (performance change).
2. Analyze key parameters such as lift, drag, thrust, airspeed changes, and bank angle changes.

Test Conditions

- Aircraft Parameters:
 - MTOM: 9163 kg
 - Wing Area: 34.4 m²
 - Wing Span: 17.17 m
 - Oswald Efficiency Factor: [0.1, 0.9, 1]
- Atmospheric Conditions:
 - Sea Level Density: 1.225 kg/m³
 - Temperature Gradient: -0.0065 K/m
 - Gas Constant: 287.05 J/(kg·K)
 - Temperature at Sea Level: 288.15 K
- Initial Parameters:
 - Altitude Range: [0, 3000, 4000, 20000] ft
 - Mach Numbers: [0, 0.4, 2]
 - Initial Angle Guess: 2°

Test Procedure

1. Pre-flight checks:

- Checking the correct functioning of measuring instruments;
- Check the correct functioning of the engines and avionics..

2. Test Execution:

a. Stabilization of the aircraft at altitude:

- I. 3000 ft at Mach= (0 - 0.4 - 2), $e = (0.1 - 0.9 - 1)$, $\varphi = (0 - 8)$;
- II. 4000 ft at Mach= (0 - 0.4 - 2), $e = (0.1 - 0.9 - 1)$, $\varphi = (0 - 8)$;
- III. 20 000 ft at Mach= (0 - 0.4 - 2), $e = (0.1 - 0.9 - 1)$, $\varphi = (0 - 8)$.

b. Record:

- I. Lift;
- II. Drag;
- III. Thrust;
- IV. V_{punto} ;
- V. Γ_{punto} ;
- VI. α_T ;
- VII. Forze aerodinamiche al variare di φ .

Safety Notes

1. Abort test in case of abnormal engine temperature indications.
2. Maintain continuous communication with air traffic control.

Test Results (Post-flight)

Observations: [_____]

Data Summary: [_____]

Issue Identified: [_____]

Approvals

Test Engineers: [_____ - _____]

Pilot in Command: [_____]

Data measured on matlab

Altitudine_ft	Mach	alphaT	v_punto	gamma_punto	e	L	D	T
0	0	0.31416	0	NaN	0.1	89858	0	0
0	0.4	-0.023076	0	0	0.1	90126	11632	11635
0	2	-0.067657	0	0	0.1	96495	97950	98174
3000	0	0.31416	0	NaN	0.1	89858	0	0
3000	0.4	-0.017704	0	0	0.1	90072	12111	12113
3000	2	-0.067443	0	0	0.1	95792	87859	88059
4000	0	0.31416	0	NaN	0.1	89858	0	0
4000	0.4	-0.015752	0	0	0.1	90052	12305	12307
4000	2	-0.067365	0	0	0.1	95572	84692	84884
20000	0	0.31416	0	NaN	0.1	89858	0	0
20000	0.4	0.031073	0	0	0.1	89289	18300	18309
20000	2	-0.065478	0	0	0.1	92846	45563	45661
0	0	0.31416	0	NaN	0.9	89858	0	0
0	0.4	-0.023158	0	0	0.9	89968	4759.5	4760.8
0	2	-0.067658	0	0	0.9	96474	97635	97859
3000	0	0.31416	0	NaN	0.9	89858	0	0
3000	0.4	-0.017782	0	0	0.9	89937	4452.8	4453.5
3000	2	-0.067443	0	-2.3579e-18	0.9	95769	87512	87712
4000	0	0.31416	0	NaN	0.9	89858	0	0
4000	0.4	-0.015827	0	0	0.9	89927	4361.4	4362
4000	2	-0.067365	0	0	0.9	95548	84334	84526
20000	0	0.31416	0	NaN	0.9	89858	0	0
20000	0.4	0.031588	0	0	0.9	89743	3646.7	3648.5
20000	2	-0.06548	0	0	0.9	92804	44928	45025
0	0	0.31416	0	NaN	1	89858	0	0
0	0.4	-0.023159	0	0	1	89966	4673.9	4675.1
0	2	-0.067658	0	0	1	96473	97631	97855
3000	0	0.31416	0	NaN	1	89858	0	0
3000	0.4	-0.017783	0	0	1	89935	4357.4	4358.1
3000	2	-0.067443	0	0	1	95769	87508	87707
4000	0	0.31416	0	NaN	1	89858	0	0
4000	0.4	-0.015828	0	0	1	89925	4262.4	4262.9
4000	2	-0.067365	0	0	1	95547	84329	84521
20000	0	0.31416	0	NaN	1	89858	0	0
20000	0.4	0.031595	0	0	1	89748	3461.6	3463.4
20000	2	-0.06548	0	0	1	92803	44920	45017

Data measured on the simulator

Altitudine_ft	Mach	alphaT	v_punto	gamma_punto	e	L	D	T
0	0				0.1			
0	0.4				0.1			
0	2				0.1			
3000	0				0.1			
3000	0.4				0.1			
3000	2				0.1			
4000	0				0.1			
4000	0.4				0.1			
4000	2				0.1			
20000	0				0.1			
20000	0.4				0.1			
20000	2				0.1			
0	0				0.9			
0	0.4				0.9			
0	2				0.9			
3000	0				0.9			
3000	0.4				0.9			
3000	2				0.9			
4000	0				0.9			
4000	0.4				0.9			
4000	2				0.9			
20000	0				0.9			
20000	0.4				0.9			
20000	2				0.9			
0	0				1			
0	0.4				1			
0	2				1			
3000	0				1			
3000	0.4				1			
3000	2				1			
4000	0				1			
4000	0.4				1			
4000	2				1			
20000	0				1			
20000	0.4				1			
20000	2				1			

SIMPLIFYING ASSUMPTIONS THAT COULD CAUSE DATA PROBLEMS

It is assumed that:

- the jet engine is ideal;
- it consumes a constant amount of fuel over time;
- a linear temperature gradient is considered, so λ may not be realistic at high altitudes;
- the simulator assumes a two-dimensional flow configuration (airfoils), without considering three-dimensional effects of the complete aircraft;
- the aircraft is a rigid body.

MAIN OBSERVATIONS

The Oswald factor directly regulates the aerodynamic efficiency of the wing: a higher value results in lower induced drag. In this regard, we speak of an elliptical planform wing, with $e=1$, which ensures an elliptical lift distribution across the wing. When $e=1$, the configuration is optimal, with low drag D and sufficient thrust T to balance the drag.

Total drag D decreases significantly as e increases, particularly noticeable at lower Mach speeds.

Furthermore, as Mach increases, lift increases slightly, unlike drag, which increases suddenly.

At high altitudes (20,000 ft), lift decreases.

Thrust T must be at least equal to total drag D to ensure hovering flight.

$\dot{v}$ is zero thanks to the conditions of equilibrium and level flight.

$\dot{\gamma}$ has values close to zero thanks to minimal variations in the angle of climb, compatible with quasi-stationary flight.

The induced drag $C_{D,i} = \frac{C_L^2}{\pi \lambda e}$ is inversely proportional to e , so if this is low, induced drag increases, worsening overall aerodynamic efficiency. Increasing the Oswald factor e reduces $C_{D,i}$ and consequently also the thrust required for flight.

GENERAL CONCLUSIONS

The Oswald factor e plays a crucial role in determining performance: with $e=0.9$ and $e=1$, efficient flight is guaranteed, with large safety margins for thrust T compared to drag D .

For $e=0.1$, performance deteriorates: the aerodynamic forces are less balanced and require thrust nearly equal to drag, with complications for stability.

PERFORMANCE ANALYSIS: VARIATION OF φ

This change in performance occurs assuming a non-symmetrical deflection of the ailerons, resulting in a bank angle φ .

Data measured on matlab

Altitudine_ft	Mach	alphaT	v_punto	gamma_punto	e	L	D	T
0	0	0.31416	0	NaN	0.1	90741	0	0
0	0.4	-0.02262	0	0	0.1	91008	11784	11787
0	2	-0.067639	0	0	0.1	97377	97956	98181
3000	0	0.31416	0	NaN	0.1	90741	0	0
3000	0.4	-0.017195	0	0	0.1	90952	12280	12282
3000	2	-0.067422	0	0	0.1	96674	87866	88066
4000	0	0.31416	0	NaN	0.1	90741	0	0
4000	0.4	-0.015224	0	0	0.1	90931	12480	12482
4000	2	-0.067344	0	0	0.1	96454	84699	84892
20000	0	0.31416	0	NaN	0.1	90741	0	0
20000	0.4	0.032045	0	0	0.1	90144	18618	18628
20000	2	-0.065438	0	0	0.1	93728	45577	45674
0	0	0.31416	0	NaN	0.9	90741	0	0
0	0.4	-0.022701	0	0	0.9	90849	4776.3	4777.6
0	2	-0.067639	0	0	0.9	97355	97635	97859
3000	0	0.31416	0	NaN	0.9	90741	0	0
3000	0.4	-0.017272	0	0	0.9	90818	4471.6	4472.3
3000	2	-0.067423	0	0	0.9	96650	87513	87712
4000	0	0.31416	0	NaN	0.9	90741	0	0
4000	0.4	-0.015298	0	0	0.9	90808	4380.9	4381.4
4000	2	-0.067344	0	0	0.9	96429	84335	84526
20000	0	0.31416	0	NaN	0.9	90741	0	0
20000	0.4	0.032587	0	0	0.9	90621	3683.1	3685.1
20000	2	-0.06544	0	0	0.9	93685	44930	45026
0	0	0.31416	0	NaN	1	90741	0	0
0	0.4	-0.022703	0	0	1	90847	4689.1	4690.3
0	2	-0.067639	0	0	1	97355	97631	97855
3000	0	0.31416	0	NaN	1	90741	0	0
3000	0.4	-0.017273	0	0	1	90817	4374.3	4375
3000	2	-0.067423	0	0	1	96650	87509	87708
4000	0	0.31416	0	NaN	1	90741	0	0
4000	0.4	-0.015299	0	0	1	90806	4280	4280.5
4000	2	-0.067344	0	0	1	96429	84330	84522
20000	0	0.31416	0	NaN	1	90741	0	0
20000	0.4	0.032594	0	0	1	90627	3494.4	3496.3
20000	2	-0.06544	0	0	1	93685	44922	45018

It is observed that varying the bank angle results in an overall increase in aerodynamic forces, thrust, and a change in the angle of attack.

Data measured on the simulator

Altitudine_ft	Mach	alphaT	v_punto	gamma_punto	e	L	D	T
0	0				0.1			
0	0.4				0.1			
0	2				0.1			
3000	0				0.1			
3000	0.4				0.1			
3000	2				0.1			
4000	0				0.1			
4000	0.4				0.1			
4000	2				0.1			
20000	0				0.1			
20000	0.4				0.1			
20000	2				0.1			
0	0				0.9			
0	0.4				0.9			
0	2				0.9			
3000	0				0.9			
3000	0.4				0.9			
3000	2				0.9			
4000	0				0.9			
4000	0.4				0.9			
4000	2				0.9			
20000	0				0.9			
20000	0.4				0.9			
20000	2				0.9			
0	0				1			
0	0.4				1			
0	2				1			
3000	0				1			
3000	0.4				1			
3000	2				1			
4000	0				1			
4000	0.4				1			
4000	2				1			
20000	0				1			
20000	0.4				1			
20000	2				1			